



中山大學管理學院
SUN YAT-SEN BUSINESS SCHOOL
SYSBS

How Aviation and Maritime Research Connect or Differ from Each Other?

- A Visualized Analysis Using CiteSpace

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Outline

- 01 Knowledge Mapping Using CiteSpace**
- 02 Research Method**
- 03 Knowledge Mapping Results**
- 04 Summary**



Outline

01 **Knowledge Mapping Using CiteSpace**

02 Research Method

03 Knowledge Mapping Results

04 Summary

Why use CiteSpace?

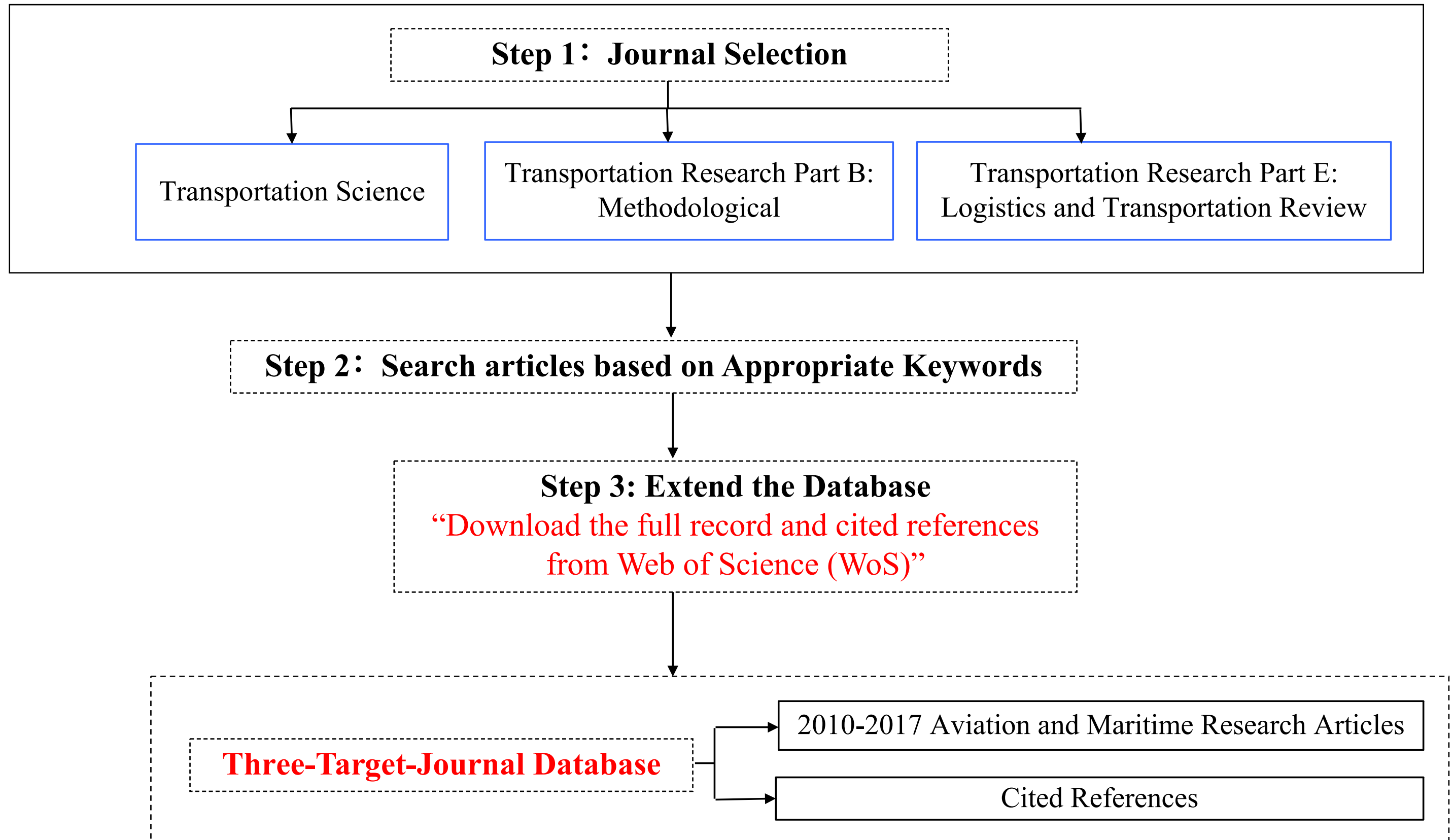
- First, using CiteSpace, major paths of aviation and maritime knowledge evolution will be revealed.
- Second, hot research topics and frontier research in the aviation and maritime field can be identified based on the frequency of popular key words used in journal articles.
- Third, authors, organizations, countries and cited journals can be clearly present.



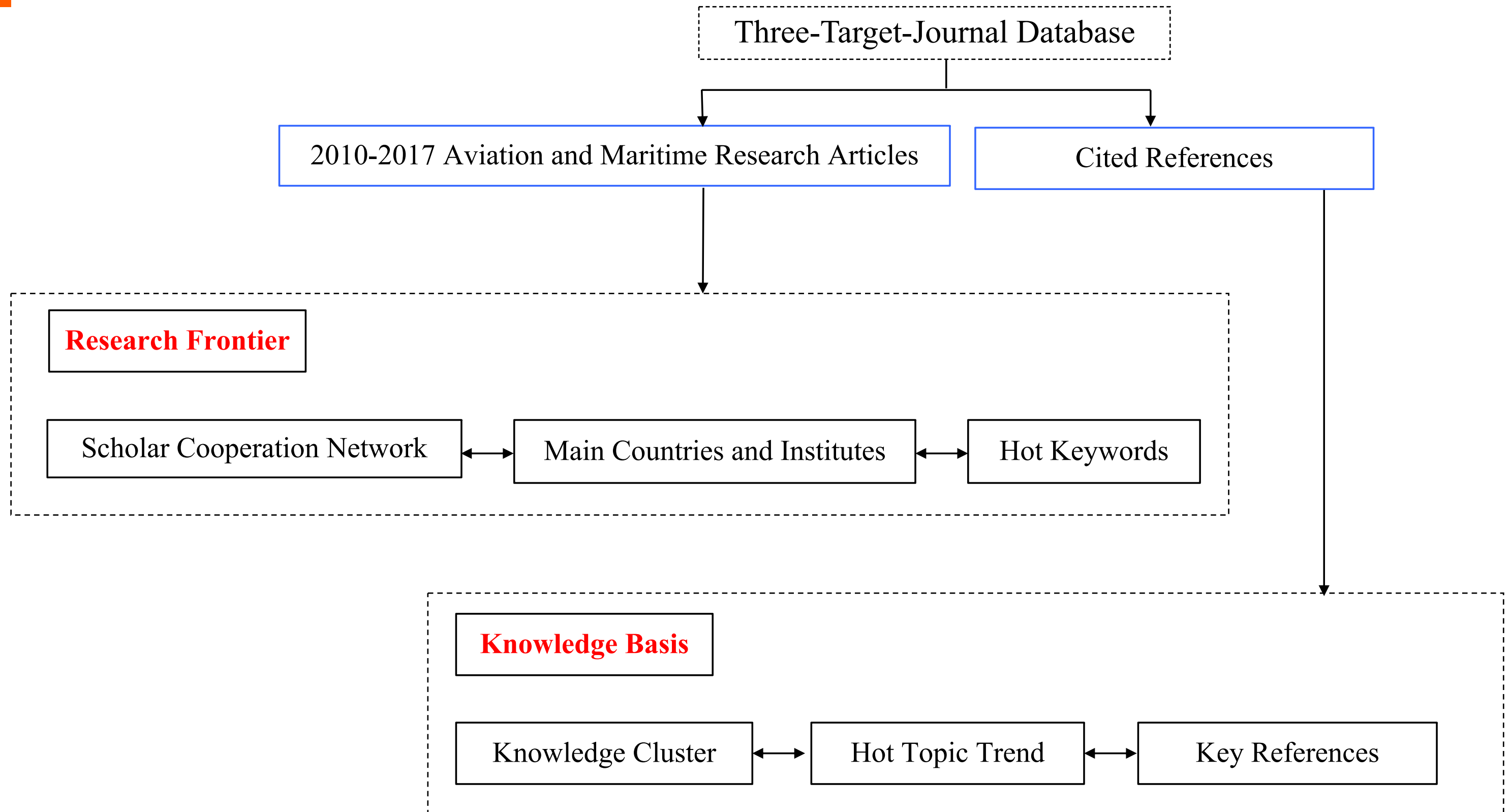
Outline

- 01 Knowledge Mapping Using CiteSpace
- 02 **Research Method**
- 03 Knowledge Mapping Results
- 04 Summary

Database Construction for Aviation and Maritime Research



Database Analysis



Keywords Selection Process

Table 1 Keywords Selection in **Aviation** Research

2010-2017 Aviation	Transportation Research Part B: Methodological		Transportation Science		Transportation Research Part E: Logistics and Transportation Review	
Air* (air/airplane/airline/airport/aircraft)	68		64		165	
Aviation	68	+0	64	+0	166	+1
Flight	70	+2	65	+1	166	+0
Plane	84	+14	76	+11	166	+0
Total	84	-	76	-	166	-

Table 2 Keywords Selection in **Maritime** Research

2010-2017 Maritime	Transportation Research Part B: Methodological		Transportation Science		Transportation Research Part E: Logistics and Transportation Review	
Ship* (ship/shipping/shipment/ shipper)	51		32		175	
Port	62	+11	44	+12	232	+57
Maritime	62	+0	45	+1	241	+9
Quay	62	+0	47	+2	245	+4
Berth	64	+2	48	+1	248	+3
Container	66	+2	53	+5	266	+18
Seaport	66	+0	53	+0	267	+1
Vessel	66	+0	53	+0	268	+1
Ocean	67	+1	53	+0	270	+2
Crane	67	+0	54	+1	271	+1
Carrier	100	+33	71	17	342	+71
Bulk	101	+1	72	+1	342	+0
Yard	103	+3	75	+3	343	+1
Total	103	-	75	-	343	-

Data collection date: August 22 2017

Aviation and Maritime Research Trend

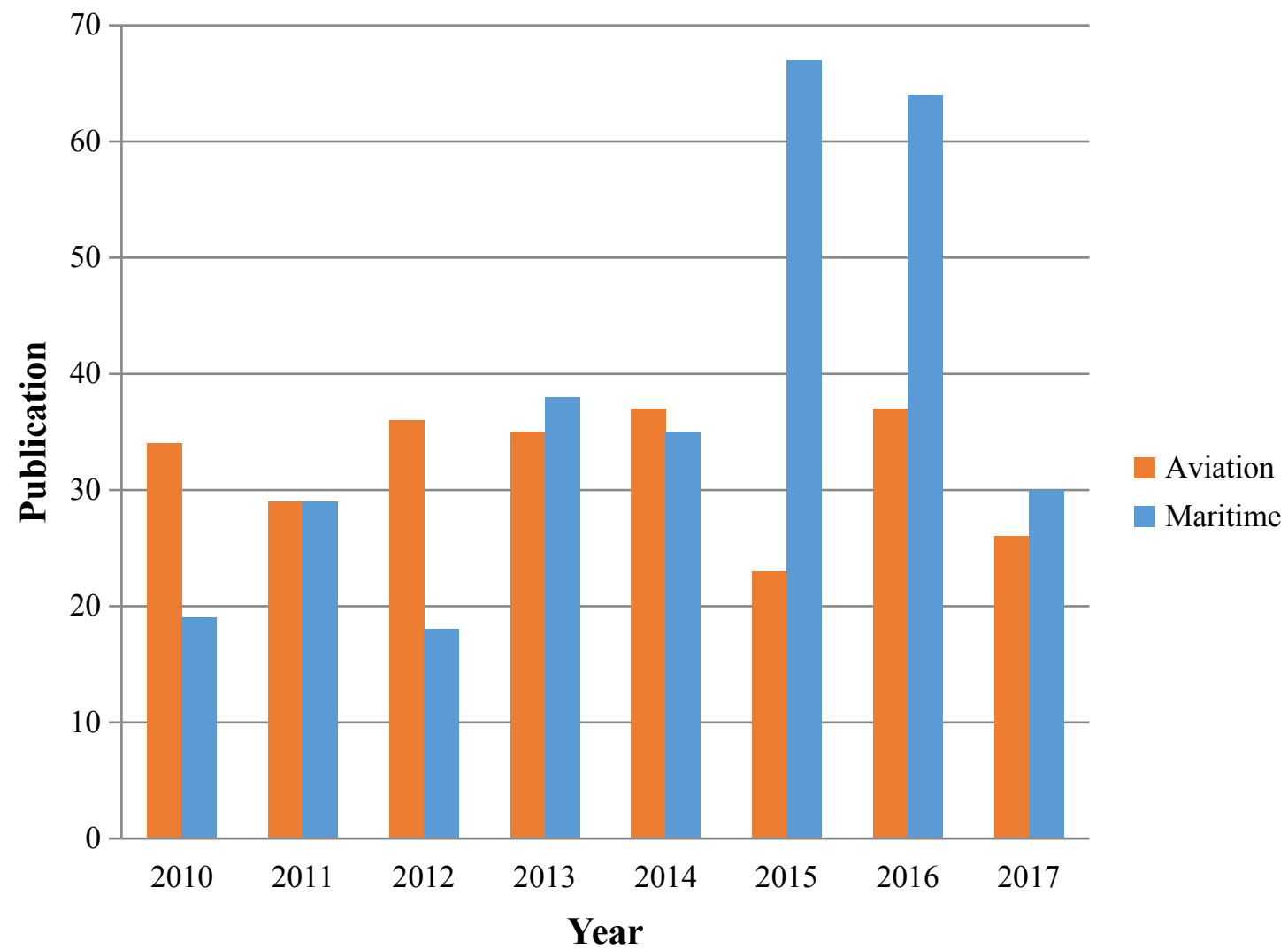


Fig .1 Publications of aviation /maritime articles in 2010–2017

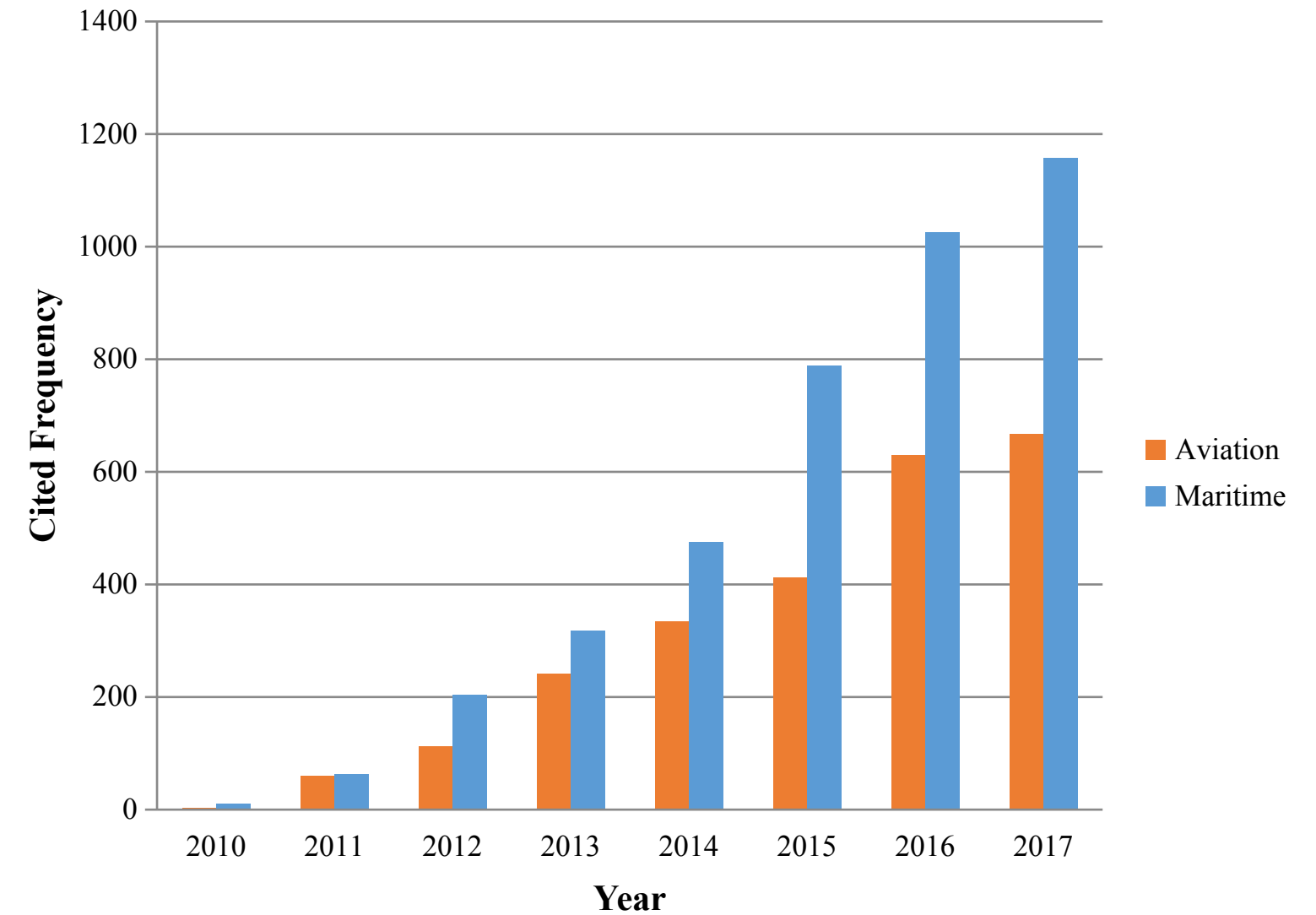


Fig.2 Cited frequency of aviation /maritime articles in 2010–2017



Outline

01 Knowledge Mapping Using CiteSpace

02 Research Method

03 **Knowledge Mapping Results**

04 Summary

Basic Setting

- Time slice starting from 2010 to 2017.
- Term source = Title , Abstract, Author, Keywords and Keywords Plus
- Node type = Cited Reference/Country /Institution /Author /Keyword
- Select 50 most cited articles per slice



3.1

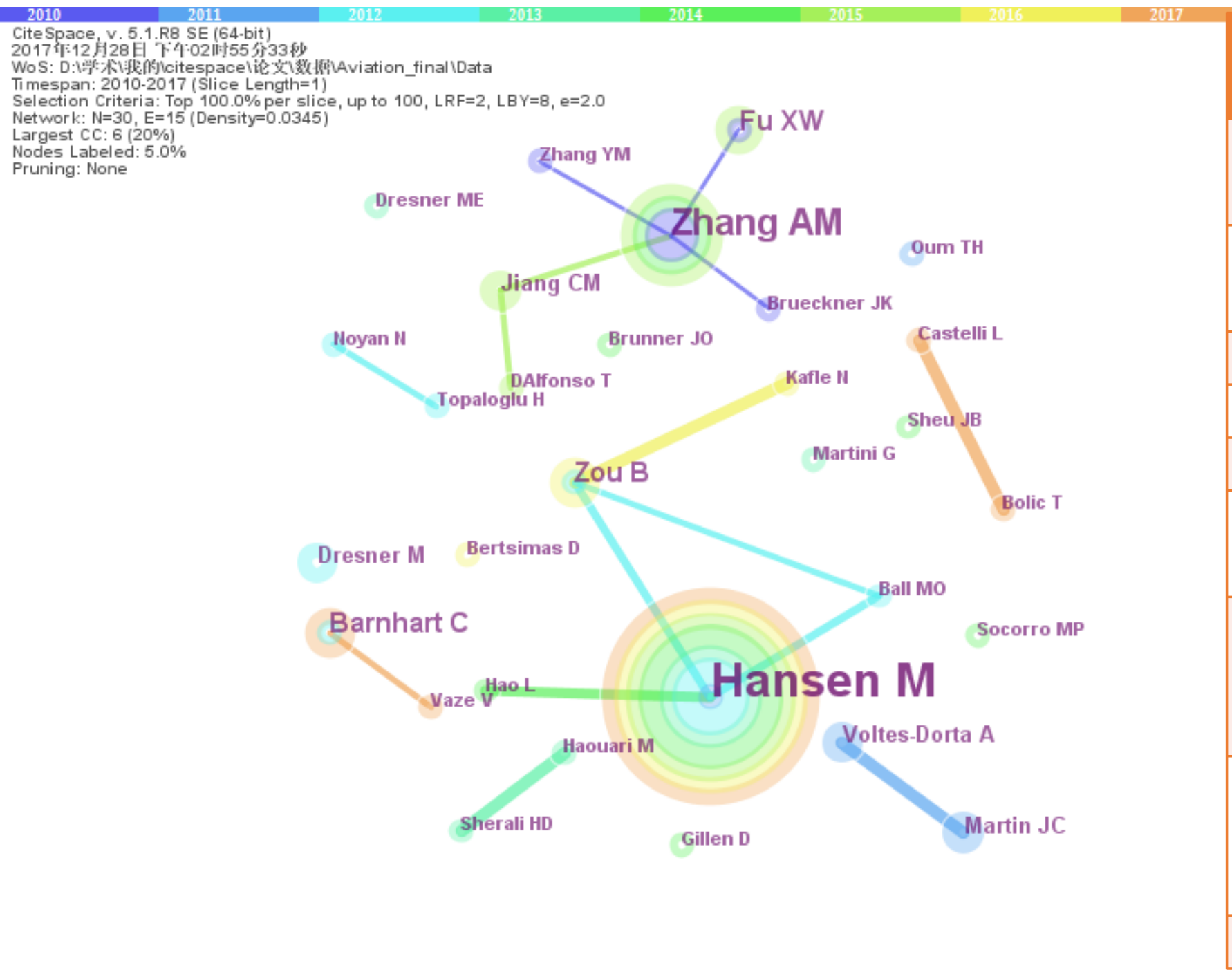
2010-2017 Aviation and Maritime Research Articles

3.2

Cited References of Selected Articles

3.1.1 Aviation Research

Scholar cooperation network



Author	Institution	Count	Centralit y	Year Begin
Hansen M	Univ Calif Berkeley	17	0.01	2011
Zhang AM	Univ British Columbia	8	0.04	2010
Fu XW	Univ Sydney	4	0.00	2010
Zou B	Univ Illinois	4	0.01	2012
Barnhart C	MIT	3	0.00	2012
Jiang CM	Univ British Columbia	3	0.02	2015
Martin JC	Univ Las Palmas Gran Canaria	3	0.00	2011
Voltes-Dorta A	Univ Las Palmas Gran Canaria	3	0.00	2011
Dresner M	Univ Maryland	3	0.00	2012

Fig. 3 Mapping of Core Scholar Cooperation Network on Aviation Research

Table 3 Core Authors in Aviation Research (Top 9)

3.1.1Aviation Research Situation

Main Countries/Regions

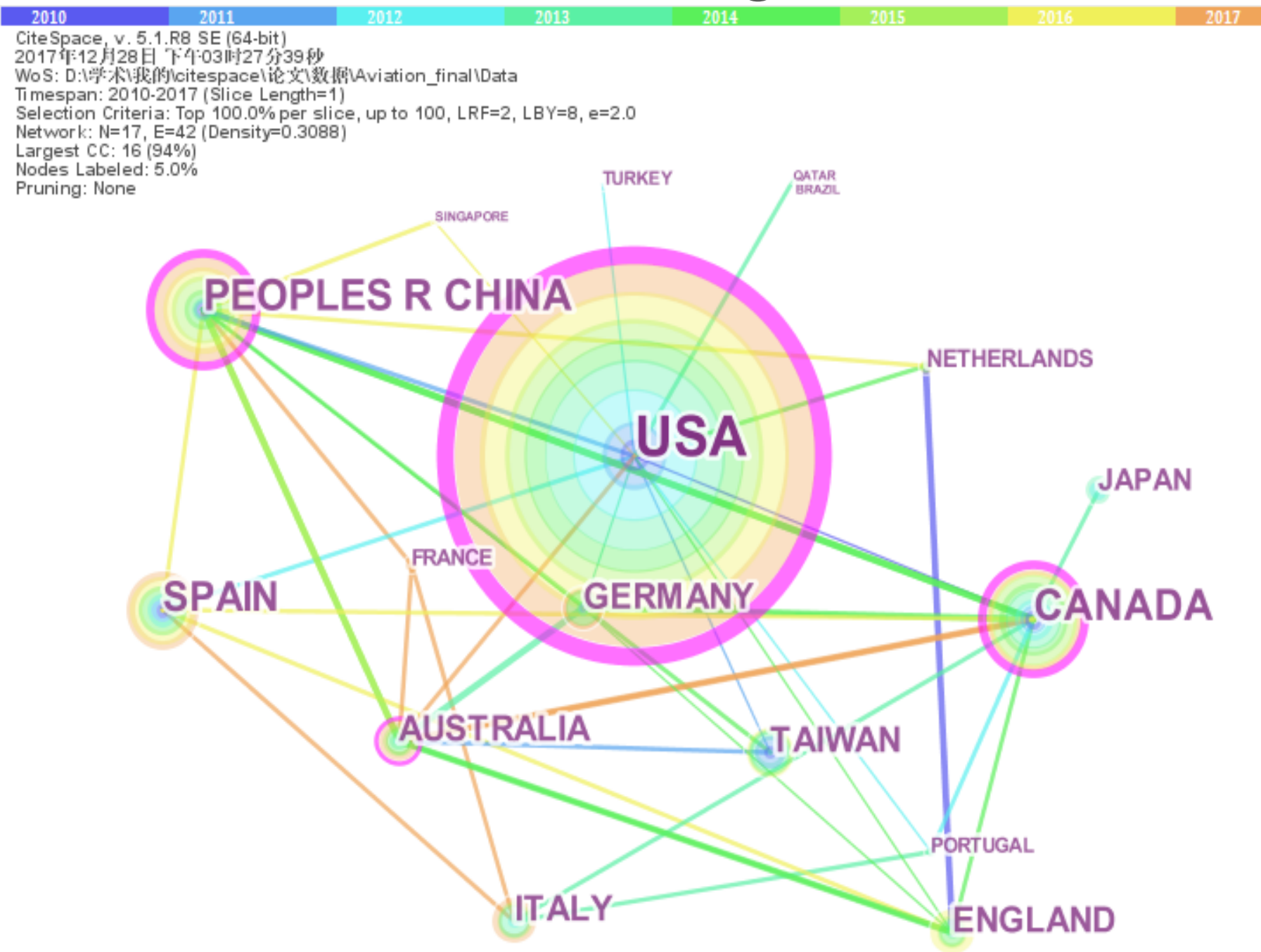


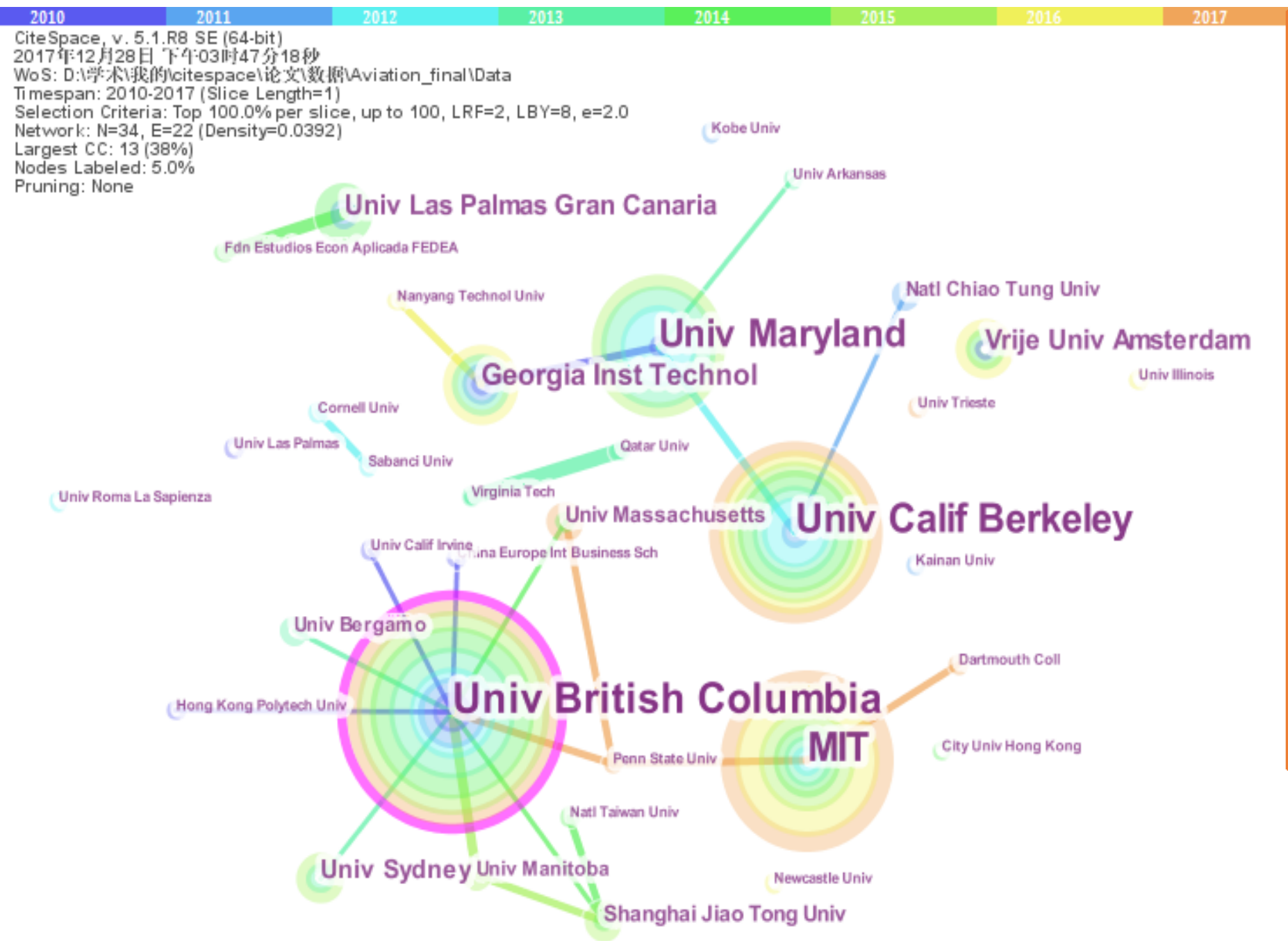
Fig. 4 Mapping of Countries in Aviation Research

Country/Region	Count	Centrality	Year Begin
USA	105	0.72	2010
Peoples R China	32	0.24	2010
Canada	30	0.36	2010
Spain	23	0.05	2010
Taiwan	17	0.00	2010
Australia	15	0.10	2011
England	15	0.04	2010
Germany	15	0.00	2010
Italy	15	0.04	2010
Japan	9	0.00	2011
Netherlands	6	0.00	2010
France	5	0.02	2016
Portugal	4	0.02	2012
Turkey	3	0.00	2012
Brazil	2	0.00	2016
Qatar	2	0.00	2013
Singapore	2	0.00	2016

Table 4 Detail of Country Connection in Aviation Research

3.1.1 Aviation Research

Main Institutions



University	Count	Centrality	Year Begin
Univ British Columbia	22	0.22	2010
Univ Calif Berkeley	18	0.02	2011
MIT	18	0.04	2012
Univ Maryland	14	0.03	2010
Georgia Inst Technol	8	0.02	2010
Vrije Univ Amsterdam	6	0.00	2011
Univ Las Palmas Gran Canaria	6	0.00	2010
Univ Sydney	5	0.00	2013
Univ Massachussetts	4	0.00	2014
Shanghai Jiao Tong Univ	4	0.04	2014

Fig.5 Mapping of Institutions in Aviation Research

Table 5 Detail of Institution Connection in Aviation Research

3.1.1 Aviation Research

Hot Keywords

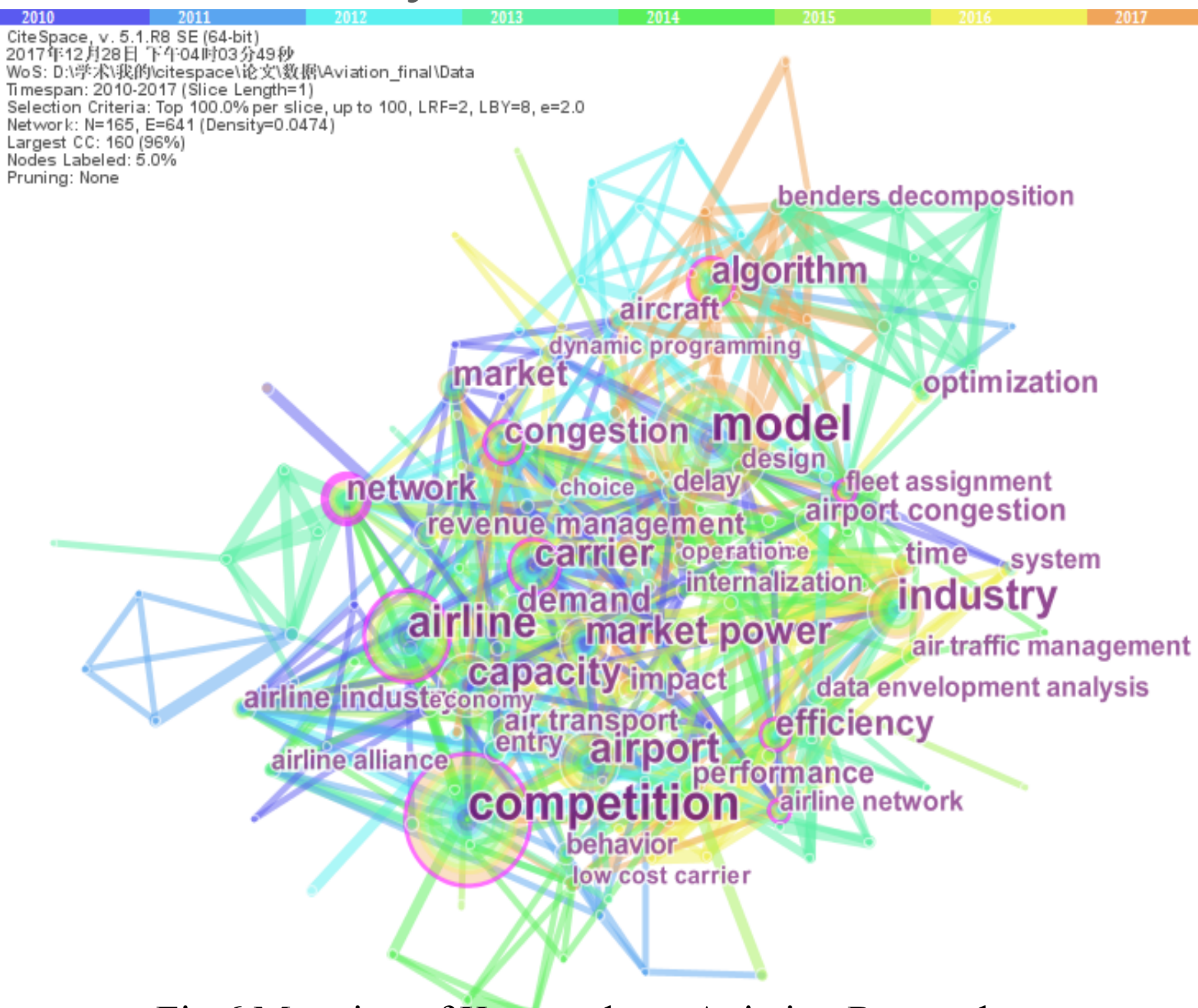


Fig.6 Mapping of Keywords on Aviation Research

Keyword	Count	Centrality	Year Begin
Competition	50	0.12	2010
Model	50	0.05	2010
Airline	36	0.19	2010
Airport	29	0.06	2010
Industry	28	0.12	2010
Capacity	27	0.05	2010
Market power	26	0.04	2010
Carrier	24	0.15	2010
Algorithm	22	0.09	2011
Congestion	18	0.22	2010
Network	17	0.24	2010
Market	17	0.02	2010
Demand	16	0.01	2011
Efficiency	15	0.12	2011
Aircraft	12	0.08	2010
Performance	12	0.08	2012
Air transport	11	0.05	2012
Revenue management	11	0.05	2010
Optimization	11	0.01	2011
Airline industry	10	0.06	2010
Time	10	0.05	2014
Impact	10	0.03	2014
Airport congestion	10	0.03	2010

Table 6 Detail of Keywords Connection in Aviation Research

3.1.2 Maritime Research

Scholar Cooperation Network

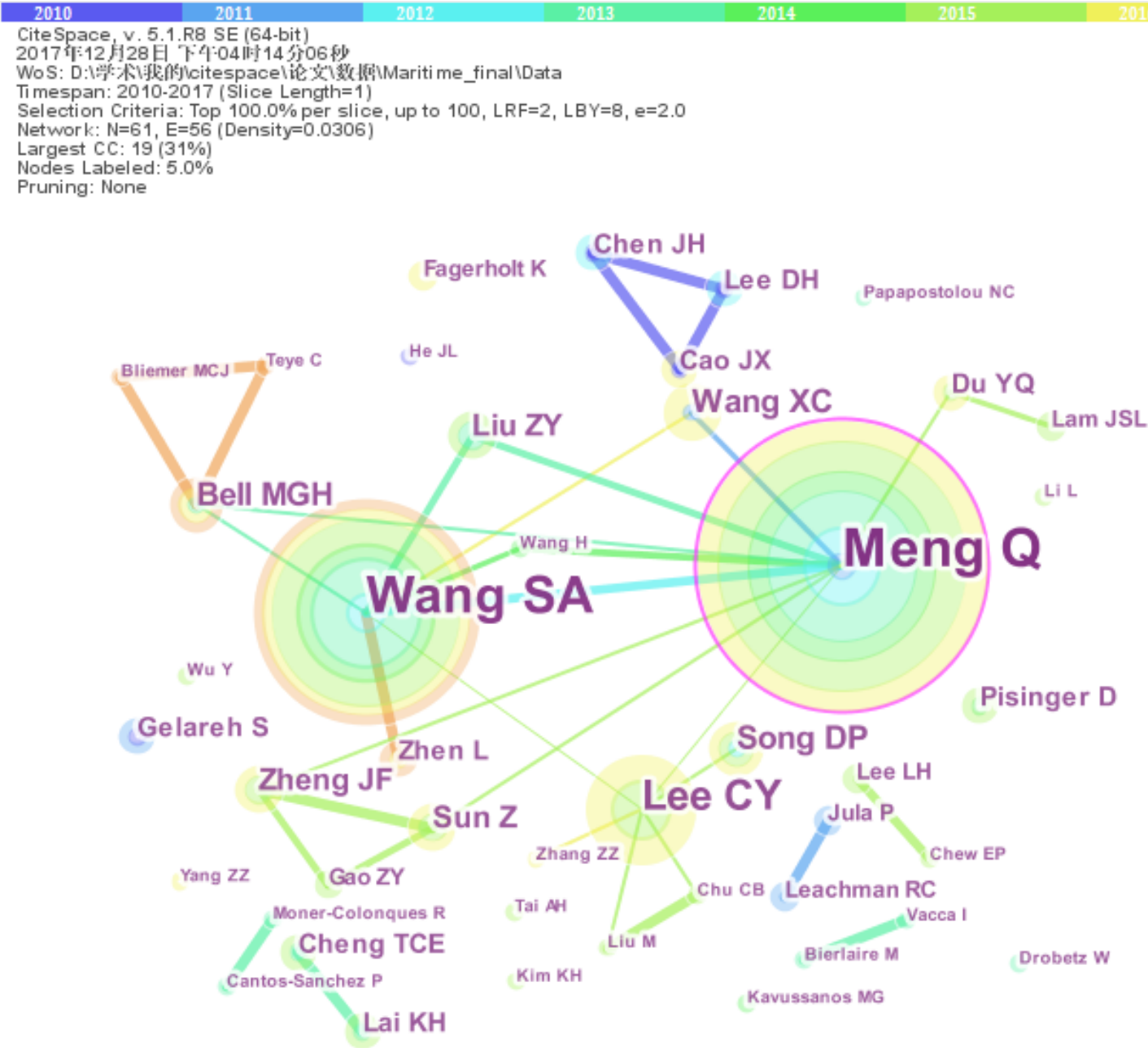


Fig. 7 Mapping of Core Scholar Cooperation Network in Maritime Research

Author	Institution	Count	Centralit y	Year Begin
Meng Q	Natl Univ Singapore	29	0.10	2011
Wang SA	Hong Kong Polytech Univ	23	0.04	2012
Lee CY	Hong Kong Univ Sci & Technol	11	0.07	2015
Wang XC	Mississippi State Univ	6	0.00	2011
Song DP	Univ Liverpool	6	0.00	2012
Bell MGH	Univ Sydney	6	0.04	2013
Sun Z	Dalian Maritime Univ	5	0.01	2015
Zheng JF	Dalian Maritime Univ	5	0.01	2015
Liu ZY	Southeast Univ	5	0.00	2013

Table 7 Core Authors in Maritime Research (Top 9)

3.1.2 Maritime Research

Main Countries/Regions

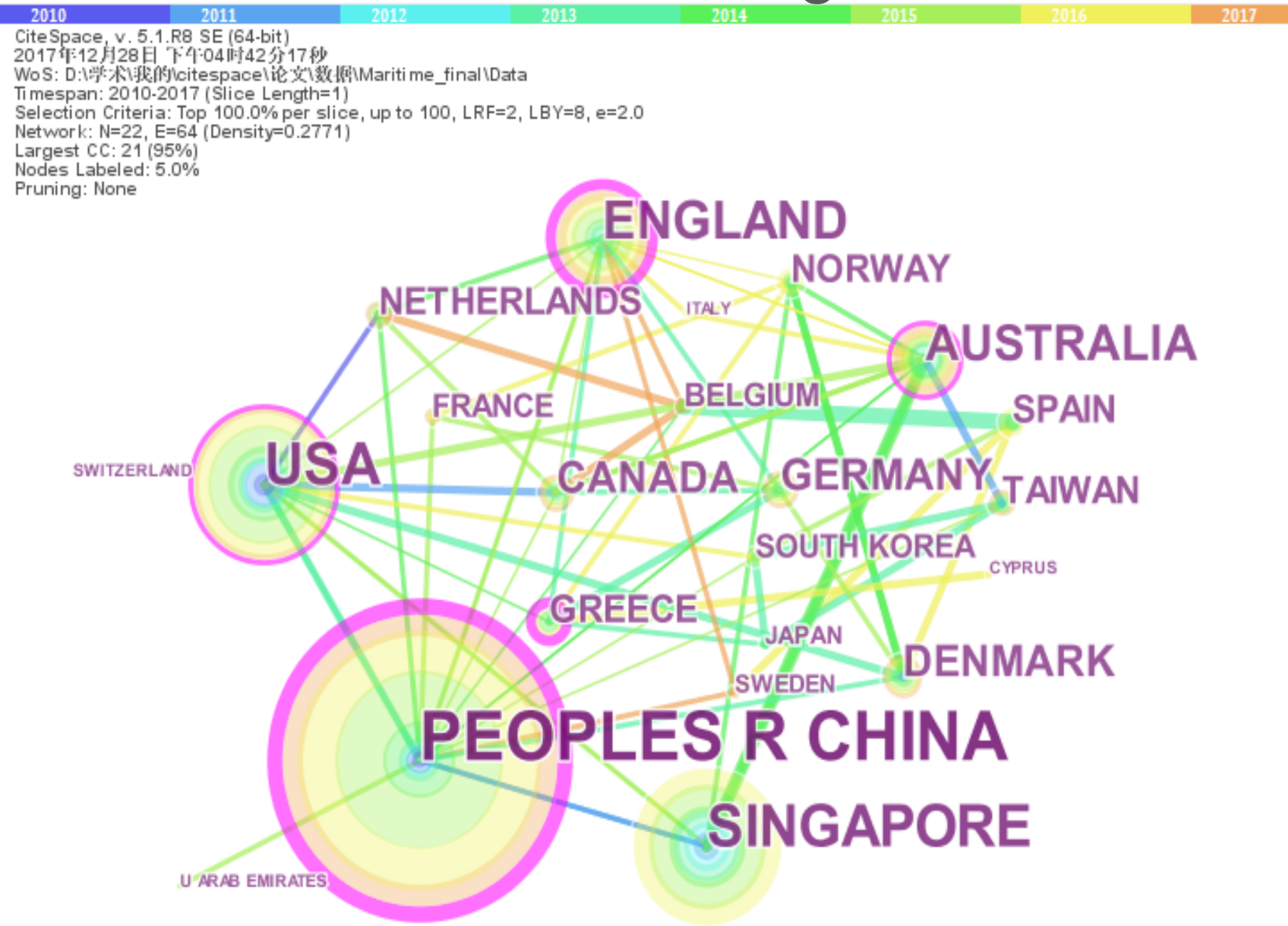


Fig.8 Mapping of Countries in Maritime Research

Country/Region	Count	Centrality	Year Begin
China	96	0.50	2010
Singapore	55	0.01	2010
USA	53	0.16	2010
England	34	0.32	2012
Australia	26	0.11	2011
Denmark	17	0.07	2011
Germany	16	0.05	2013
Canada	15	0.04	2011
Taiwan	12	0.05	2011
Spain	12	0.03	2011
Greece	11	0.26	2013
Netherlands	11	0.00	2010
Norway	10	0.08	2013
Belgium	7	0.04	2013
France	7	0.02	2015
South korea	7	0.08	2013
Sweden	4	0.02	2016
Japan	4	0.03	2013
U arab emirates	2	0.00	2015
Italy	2	0.00	2016
Cyprus	2	0.00	2016
Switzerland	2	0.00	2013

Table 8 Detail of Country Connection in Maritime Research

3.1.2 Maritime Research

Main Institutions

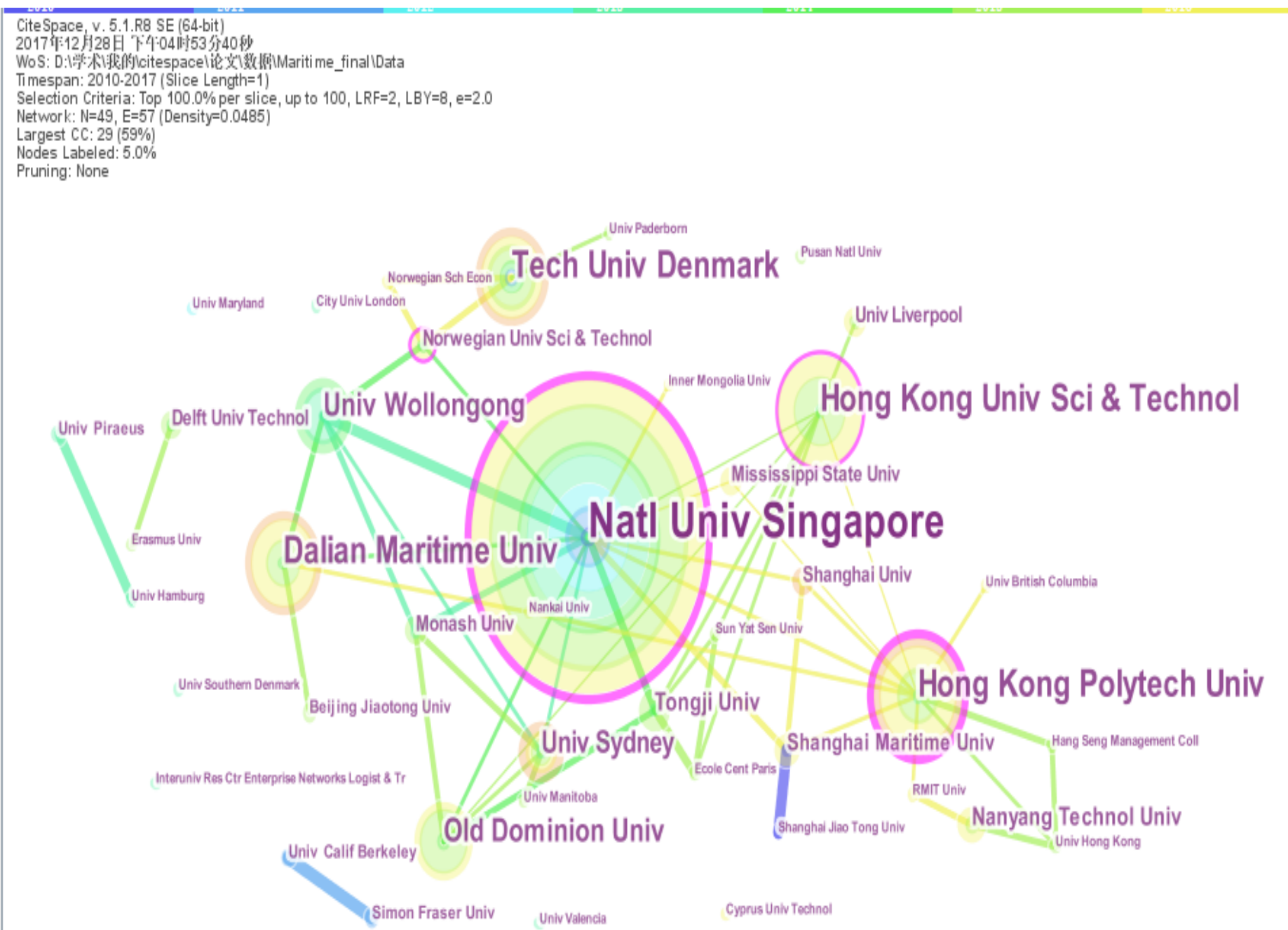


Fig.9 Mapping Of Institutions in Maritime Research

Institution	Count	Centrality	Year Begin
Natl Univ Singapore	46	0.38	2010
Hong Kong Polytech Univ	17	0.25	2013
Hong Kong Univ Sci & Technol	16	0.12	2014
Tech Univ Denmark	15	0.05	2011
Dalian Maritime Univ	15	0.06	2014
Old Dominion Univ	12	0.01	2014
Univ Wollongong	11	0.03	2013
Univ Sydney	9	0.05	2013
Tongji Univ	6	0.03	204
Nanyang Technol Univ	6	0.00	2015

Table 9 Detailed Institution Connection in Maritime Research

3.1.2 Maritime Research

Hot Keywords

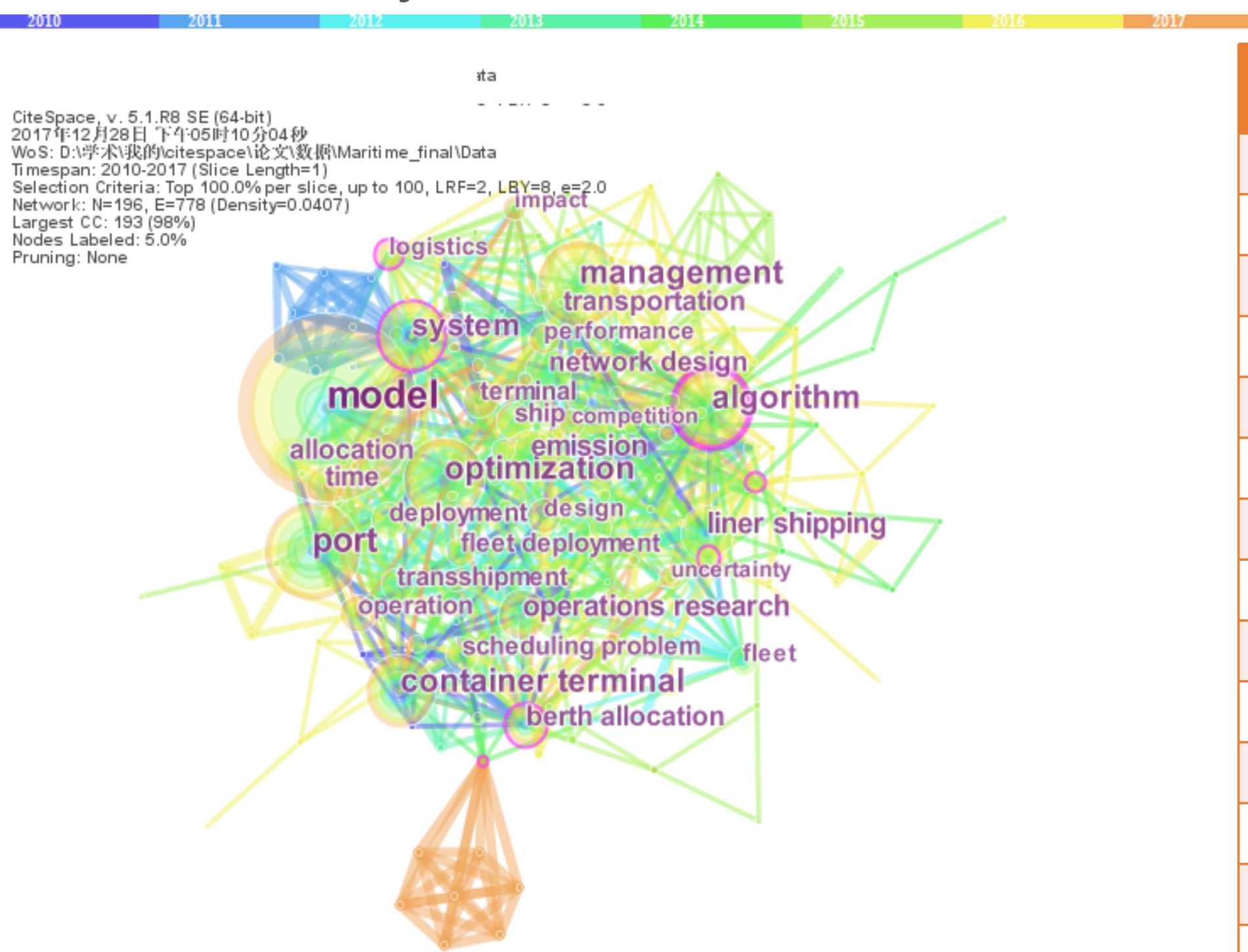


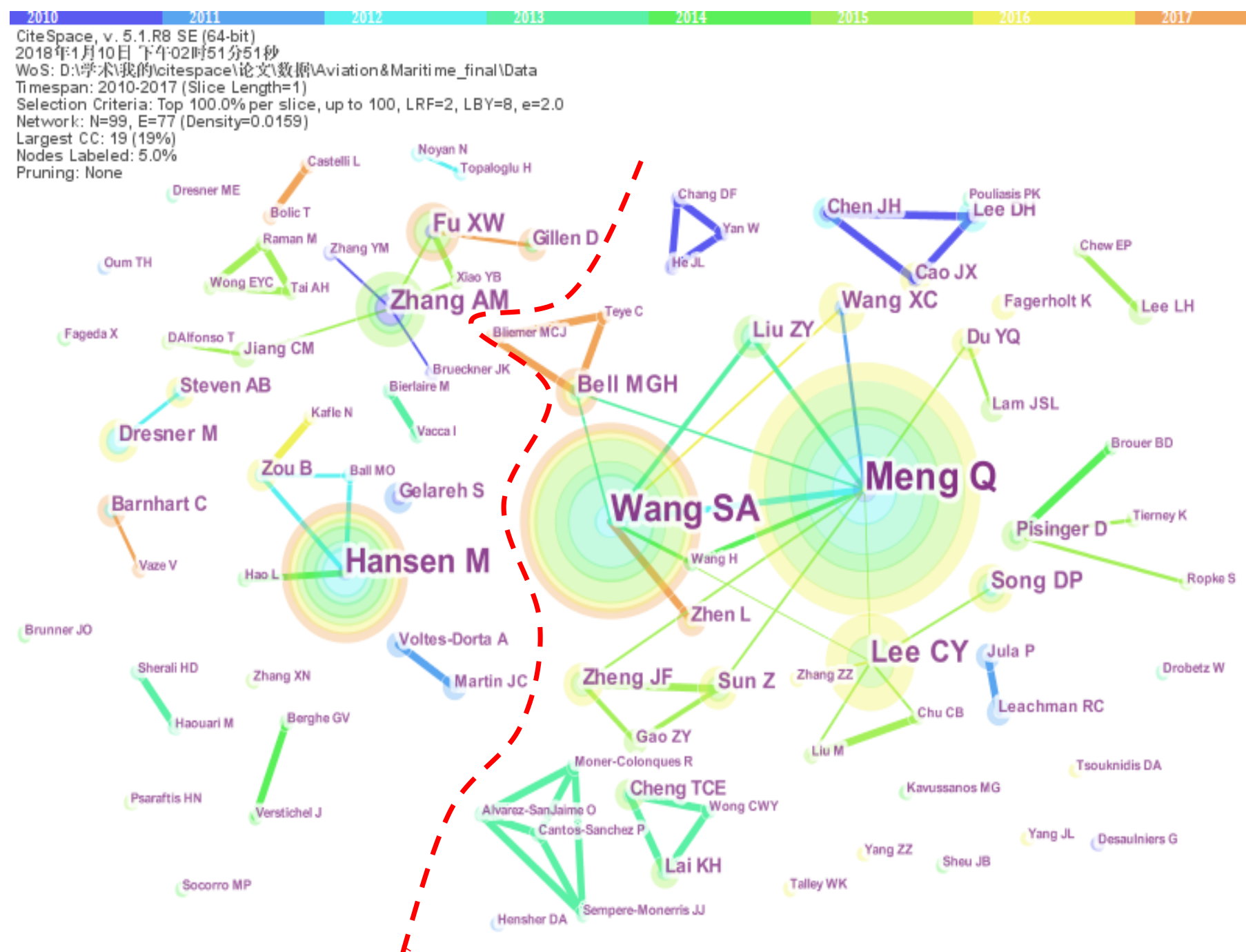
Fig.10 Mapping of Keywords in Maritime Research

Keyword	Count	Centrality	Year Begin
Model	81	0.06	2010
Port	44	0.04	2010
Management	38	0.07	2011
Optimization	38	0.03	2010
Algorithm	35	0.27	2010
Container terminal	34	0.10	2010
System	32	0.16	2010
Operations research	25	0.07	2010
Liner shipping	25	0.04	2010
Emission	23	0.10	2012
Berth allocation	21	0.14	2010
Network design	21	0.03	2012
Allocation	20	0.02	2010
Time	20	0.03	2013

Table 10 Detailed Keyword Connection in Maritime Research

3.1.3 Aviation&Maritime Research

Scholar Cooperation Network



Author	Institution	Count	Centrality	Year Begin
Meng Q	Natl Univ Singapore	29	0.04	2011
Wang SA	Hong Kong Polytech Univ	23	0.02	2012
Hansen M	Univ Calif Berkeley	17	0.00	2011
Lee CY	Hong Kong Univ Sci & Technol	11	0.03	2015
Zhang AM	Univ British Columbia	9	0.01	2010
Fu XW	Univ Sydney	7	0.00	2010
Wang XC	Mississippi State Univ	6	0.00	2011
Song DP	Univ Liverpool	6	0.00	2012
Bell MGH	Univ Sydney	6	0.01	2013
Liu ZY	Southeast Univ	5	0.00	2013
Dresner M	Univ Maryland	5	0.00	2012
Sun Z	Dalian Maritime Univ	5	0.00	2015
Zheng JF	Dalian Maritime Univ	5	0.00	2015

Fig. 11 Mapping of Core Scholar Cooperation Network in Aviation & Maritime Research

Table 11 Core Authors in Aviation & Maritime Research (Top 9)



3.1

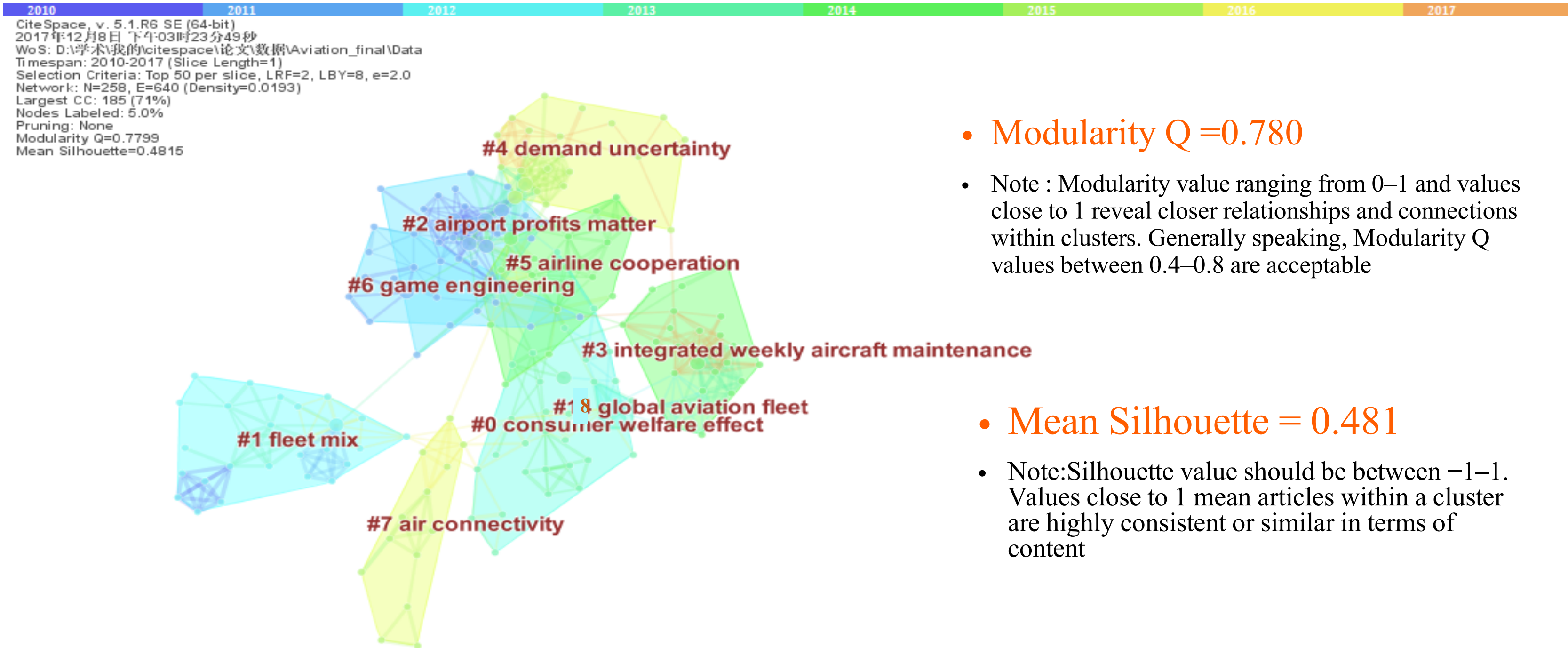
2010-2017 Aviation and Maritime Research Articles

3.2

Cited References of Selected Articles

3.2.1 Aviation Knowledge Basis

Knowledge Clusters



- **Modularity $Q = 0.780$**

- Note : Modularity value ranging from 0–1 and values close to 1 reveal closer relationships and connections within clusters. Generally speaking, Modularity Q values between 0.4–0.8 are acceptable

- **Mean Silhouette = 0.481**

- Note:Silhouette value should be between $-1-1$. Values close to 1 mean articles within a cluster are highly consistent or similar in terms of content

Fig.12 Co-citation Clusters of Cited References in Aviation Research

3.2.1 Aviation Knowledge Basis

Knowledge Clusters

Table 12 Top Terms in Each Knowledge Cluster of Aviation Research

Cluster	Size	Silhouette	Mean (year)	Top terms (LLR)
0	29	0.834	2009	consumer welfare effect (315.75, 1.0e-4); malev hungarian airline (315.75, 1.0e-4); flight frequency (197.4, 1.0e-4); comprehensive assessment (197.4, 1.0e-4); airport profits matter (43.33, 1.0e-4);
1	28	0.981	2007	fleet mix (365.73, 1.0e-4); environmental efficiency (350.88, 1.0e-4); airport quality (119.37, 1.0e-4); malmquist index decomposition assessment (119.37, 1.0e-4); productivity change (119.37, 1.0e-4);
2	28	0.772	2007	airport profits matter (451.11, 1.0e-4); pricing v (451.11, 1.0e-4); slot policies (451.11, 1.0e-4); congestion pricing (117.4, 1.0e-4); airport capacity (117.4, 1.0e-4);
3	26	0.973	2009	integrated weekly aircraft maintenance (330.42, 1.0e-4); fleet assignment problem (330.42, 1.0e-4); network-based model (330.42, 1.0e-4); demand-oriented integrated scheduling (77.71, 1.0e-4); point-to-point airline (77.71, 1.0e-4)
4	23	0.915	2011	demand uncertainty (474.74, 1.0e-4); light-handed airport regulation (355.88, 1.0e-4); airport network (101.24, 1.0e-4); slot auction (101.24, 1.0e-4); airport profits matter (41.18, 1.0e-4)
5	21	0.832	2009	airline cooperation (294.24, 1.0e-4); hub airport capacity constraint (294.24, 1.0e-4); high-speed rail (185.49, 1.0e-4); airport profits matter (40.11, 1.0e-4); pricing v (40.11, 1.0e-4);
6	13	0.869	2006	game engineering (172.49, 1.0e-4); cost-benefit analysis (172.49, 1.0e-4); air transport competition (172.49, 1.0e-4); high-speed rail (65.59, 1.0e-4); aircraft design (27.8, 1.0e-4)
7	13	0.943	2011	air connectivity (164.1, 1.0e-4); air transport (164.1, 1.0e-4); new metric (164.1, 1.0e-4); airport profits matter (16.52, 1.0e-4); pricing v (16.52, 1.0e-4)
8	4	0.992	2008	global aviation fleet (24.29, 1.0e-4); environmental efficiency (11.28, 0.001); fleet mix (1.47, 0.5); airport profits matter (1.16, 0.5); pricing v (1.16, 0.5);

3.2.1 Aviation Knowledge Basis

Knowledge Clusters

Table 13 The Most Active Citer in Each Cluster of Aviation Research

Cluster	Coverage	Author and year	Articles
0	24%	Bilotkach, V (2014)	Estimating the consumer welfare effects of de-hubbing: the case of malev hungarian airlines
	24%	Zou, B (2014)	Flight delay impact on airfare and flight frequency: a comprehensive assessment
1	25%	De Nicola, A (2013)	Airport quality and productivity changes: a malmquist index decomposition assessment
	25%	Martini, G (2013)	The influence of fleet mix, ownership and lccs on airports' technical/environmental efficiency
2	43%	Basso, LJ (2010)	Pricing vs. Slot policies when airport profits matter
	36%	Zhang, AM (2010)	Airport capacity and congestion pricing with both aeronautical and commercial operations
3	27%	Liang, Z (2013)	A network-based model for the integrated weekly aircraft maintenance routing and fleet assignment problem
	19%	Faust, O (2017)	Demand-oriented integrated scheduling for point-to-point airlines
4	30%	Yang, HJ (2015)	A comparison of price-cap and light-handed airport regulation with demand uncertainty
	26%	Sheng, D (2015)	Slot auction in an airport network with demand uncertainty
5	38%	Jiang, CM (2014)	Effects of high-speed rail and airline cooperation under hub airport capacity constraint
6	38%	Adler, N (2010)	High-speed rail and air transport competition: game engineering as tool for cost-benefit analysis
	15%	Brueckner, JK (2010)	Airline emission charges: effects on airfares, service quality, and aircraft design
7	54%	Allroggen, F (2015)	How air transport connects the world - a new metric of air connectivity and its evolution between 1990 and 2012
8	25%	Adler, N (2013)	Measuring the environmental efficiency of the global aviation fleet
	25%	Martini, G (2013)	The influence of fleet mix, ownership and lccs on airports' technical/environmental efficiency

3.2.1 Aviation Knowledge Basis

Hot Topic Trend

Top 5 References with the Strongest Citation Bursts

References	Year	Strength	Begin	End	2010 - 2017
MAYER C, 2003, AM ECON REV, V93, P1194, DOI	2003	2.2448	2010	2011	
ADLER N, 2005, TRANSPORT SCI, V39, P58, DOI	2005	2.2448	2010	2011	
PELS E, 2004, J URBAN ECON, V55, P257, DOI	2004	3.8717	2010	2012	
ZHANG AM, 2006, J URBAN ECON, V60, P229, DOI	2006	2.8328	2010	2011	
HOFER C, 2008, TRANSPORT RES E-LOG, V44, P864, DOI	2008	2.2982	2012	2014	

Table 14 Top 5 References with Strongest Citation Bursts in Aviation Research

Bursts	References	Cluster
3.87	Pels E, Verhoef E T. The economics of airport congestion pricing[J]. Journal of Urban Economics, 2004, 55(2): 257-277.	2
2.83	Zhang, A., & Zhang, Y. (2006). Airport capacity and congestion when carriers have market power. Journal of Urban Economics, 60(2), 229-247.	2
2.30	Hofer, C., Windle, R. J., & Dresner, M. E. (2008). Price premiums and low cost carrier competition. Transportation Research Part E: Logistics and Transportation Review, 44(5), 864-882.	0
2.24	Adler, N. (2005). Hub-spoke network choice under competition with an application to Western Europe. Transportation science, 39(1), 58-72.	6
2.24	Mayer, C., & Sinai, T. (2003). Network effects, congestion externalities, and air traffic delays: Or why not all delays are evil. The American Economic Review, 93(4), 1194-1215.	2

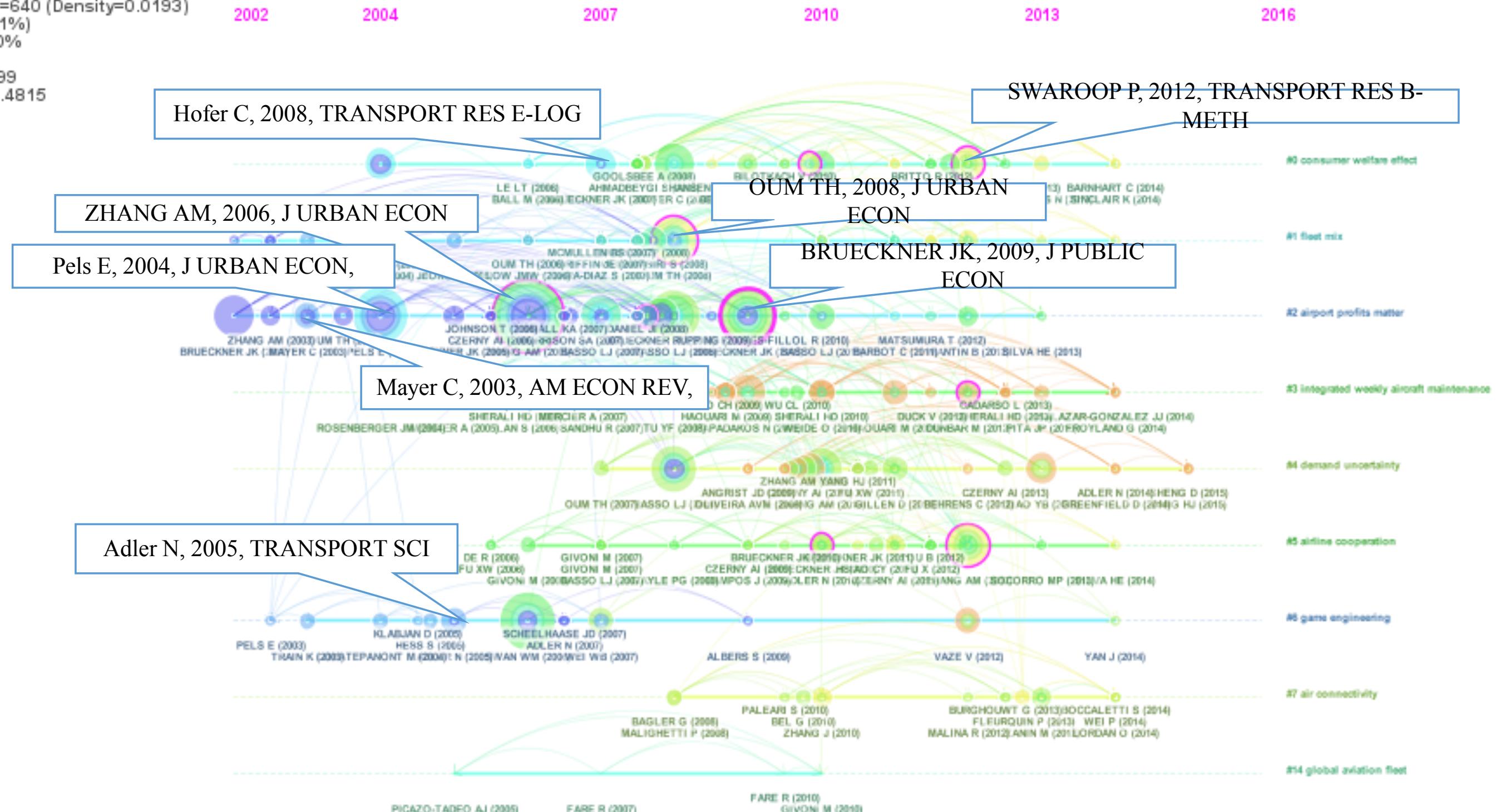
3.2.1 Aviation Knowledge Basis

Hot Topic Trend

2010 2011 2012 2013 2014 2015 2016 2017

CiteSpace, v. 5.1.R6 SE (64-bit)
2017年12月8日 下午02时45分07秒
WoS: D:\学术\我的citespace\论文\数据\Aviation_final\Data
Timespan: 2010-2017 (Slice Length=1)
Selection Criteria: Top 50 per slice, LRF=2, LBY=8, e=2.0
Network: N=258, E=640 (Density=0.0193)
Largest CC: 185 (71%)
Nodes Labeled: 5.0%
Pruning: None
Modularity Q=0.7799
Mean Silhouette=0.4815

Fig. 13 Time Line of Aviation Research



3.2.1 Aviation Knowledge Basis

Key References

Table 15 Details of the Top 10 References in Aviation Research

References	count	Centrality	Cluster
Zhang, A., & Zhang, Y. (2006). Airport capacity and congestion when carriers have market power. Journal of Urban Economics, 60(2), 229-247.	14	0.15	2
Pels E, Verhoef E T. The economics of airport congestion pricing[J]. Journal of Urban Economics, 2004, 55(2): 257-277.	11	0.01	2
Swan, W. M., & Adler, N. (2006). Aircraft trip cost parameters: A function of stage length and seat capacity. Transportation Research Part E: Logistics and Transportation Review, 42(2), 105-115.	11	0.03	6
Brueckner, J. K. (2009). Price vs. quantity-based approaches to airport congestion management. Journal of Public Economics, 93(5), 681-690.	10	0.38	2
Basso, L. J. (2008). Airport deregulation: Effects on pricing and capacity. International Journal of Industrial Organization, 26(4), 1015-1031.	10	0.05	2
Basso, L. J., & Zhang, A. (2008). On the relationship between airport pricing models. Transportation Research Part B: Methodological, 42(9), 725-735.	9	0.02	4
Oum, T. H., Yan, J., & Yu, C. (2008). Ownership forms matter for airport efficiency: A stochastic frontier investigation of worldwide airports. Journal of Urban	9	0.19	1
Hofer, C., Windle, R. J., & Dresner, M. E. (2008). Price premiums and low cost carrier competition. Transportation Research Part E: Logistics and Transportation Review, 44(5), 864-882.	8	0.00	0
Zhang, A., Fu, X., & Yang, H. G. (2010). Revenue sharing with multiple airlines and airports. Transportation Research Part B: Methodological, 44(8), 944-959.	8	0.01	4
Basso, L. J., & Zhang, A. (2007). Congestible facility rivalry in vertical structures. Journal of Urban Economics, 61(2), 218-237.	8	0.04	2

Note : Count from all clusters

3.2.1 Aviation Knowledge Basis

Most Cited Journals in Cited References

Table 16 Top 6 Most Cited Journals in the References of Aviation Research

Citation Count	Journals
35	TRANSPORT RES E-LOG
25	OPER RES
24	TRANSPORT RES B-METH
23	Tourism Management
18	TRANSPORT SCI
12	J URBAN ECON

Note : Count from all clusters

3.2.2 Maritime Knowledge Basis

Knowledge Clusters

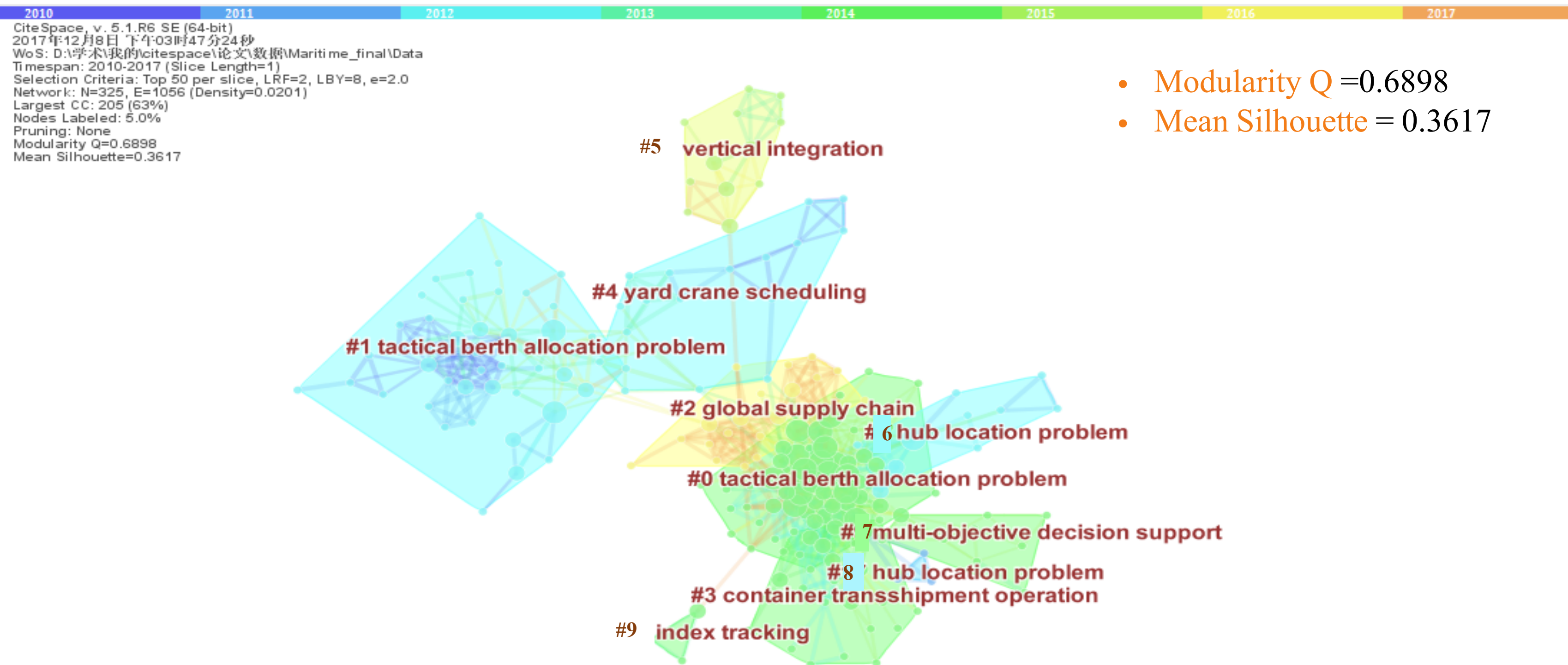


Fig.14 Co-citation Clusters of Cited References in Maritime Research

3.2.2 Maritime Knowledge Basis_

Knowledge Clusters

Table 17 Top terms in Each Knowledge Cluster in Maritime Research

Cluster	Size	Sihuoette	Mean Year	LLR
0	61	0.584	2011	tactical berth allocation problem (572.77, 1.0E-4); liner ship route schedule (466.84, 1.0E-4); robust schedule design (304.38, 1.0E-4); liner shipping service (304.38, 1.0E-4); yard crane scheduling (170.05, 1.0E-4);
1	46	0.952	2008	tactical berth allocation problem (1208.04, 1.0E-4); continuous berth allocation problem (329.83, 1.0E-4); greedy randomized adaptive search solution (329.83, 1.0E-4); liner ship route schedule (310.08, 1.0E-4); robust schedule design (151.68, 1.0E-4);
2	22	0.846	2013	global supply chain (476.07, 1.0E-4); ocean container transport (476.07, 1.0E-4); research opportunities (476.07, 1.0E-4); liner ship route schedule (187.36, 1.0E-4); robust schedule design (91.93, 1.0E-4);
3	20	0.84	2010	container transshipment operation (60.91, 1.0E-4); liner ship fleet deployment (60.91, 1.0E-4); liner ship route schedule (45.27, 1.0E-4); robust schedule design (30.38, 1.0E-4); liner shipping service (30.38, 1.0E-4);
4	18	0.958	2007	yard crane scheduling (224.56, 1.0E-4); hybrid parallel genetic algorithm (224.56, 1.0E-4); liner ship route schedule (56.67, 1.0E-4); robust schedule design (27.89, 1.0E-4); liner shipping service (27.89, 1.0E-4);
5	12	0.991	2010	vertical integration (194.16, 1.0E-4); horizontal integration (152.6, 1.0E-4); liner ship route schedule (17.65, 1.0E-4); robust schedule design (8.69, 0.005); liner shipping service (8.69, 0.005);
6	11	0.944	2007	hub location problem (29.46, 1.0E-4); transportation network (29.46, 1.0E-4); liner ship route schedule (22.59, 1.0E-4); robust schedule design (15.02, 0.001); liner shipping service (15.02, 0.001)
7	9	0.958	2010	multi-objective decision support (208.71, 1.0E-4); future direction (208.71, 1.0E-4); environmental sustainability (208.71, 1.0E-4); liner ship route schedule (40.53, 1.0E-4); robust schedule design (19.95, 1.0E-4); liner shipping service (19.95, 1.0E-4)
8	3	0.975	2005	hub location problem (10.4, 0.005); transportation network (10.4, 0.005); liner ship route schedule (0.97, 0.5); robust schedule design (0.48, 0.5); liner shipping service (0.48, 0.5)
9	3	0.991	2007	index tracking (23.57, 1.0E-4); using evolutionary algorithm (23.57, 1.0E-4); shipping stock (23.57, 1.0E-4); freight market (23.57, 1.0E-4); capital structure decision (11.67, 0.001)

3.2.2 Maritime Knowledge Basis

Knowledge Clusters

Table 18 The Most Active Citer in Each Cluster of Maritime Research

Cluster	Coverage	Author and Year	Articles
0	25%	Wang, SA (2012)	Liner ship route schedule design with sea contingency time and port time uncertainty
	20%	Wang, SA (2012)	Robust schedule design for liner shipping services
	16%	Wang, SA (2012)	Sailing speed optimization for container ships in a liner shipping network
	15%	Wang, SA (2012)	Liner ship fleet deployment with container transshipment operations
1	28%	Giallombardo, G (2010)	Modeling and solving the tactical berth allocation problem
	24%	Lee, DH (2010)	The continuous berth allocation problem: a greedy randomized adaptive search solution
2	55%	Lee, CY (2017)	Ocean container transport in global supply chains: overview and research opportunities
3	35%	Wang, SA (2012)	Liner ship route schedule design with sea contingency time and port time uncertainty
	25%	Wang, SA (2012)	Liner ship fleet deployment with container transshipment operations
4	39%	He, JL (2010)	A hybrid parallel genetic algorithm for yard crane scheduling
5	33%	Alvarez-SanJaime, O (2013)	Competition and horizontal integration in maritime freight transport
	33%	Alvarez-SanJaime, O (2013)	Vertical integration and exclusivities in maritime freight transport
6	27%	Wang, SA (2012)	Robust schedule design for liner shipping services
	18%	Gelareh, S (2011)	Hub location problems in transportation networks
7	67%	Mansouri, SA (2015)	Multi-objective decision support to enhance environmental sustainability in maritime shipping: a review and future directions
8	33%	Gelareh, S (2011)	Hub location problems in transportation networks
9	33%	Andriosopoulos, K (2013)	Portfolio optimization and index tracking for the shipping stock and freight markets using evolutionary algorithms
	33%	Drobetz, W (2013)	Capital structure decisions of globally-listed shipping companies

Hot Topic Trend

Table 19 Top 10 References with Strongest Citation Bursts in Maritime Research

[illegible]

3.2.2 Maritime Knowledge Basis

Hot Topic Trend

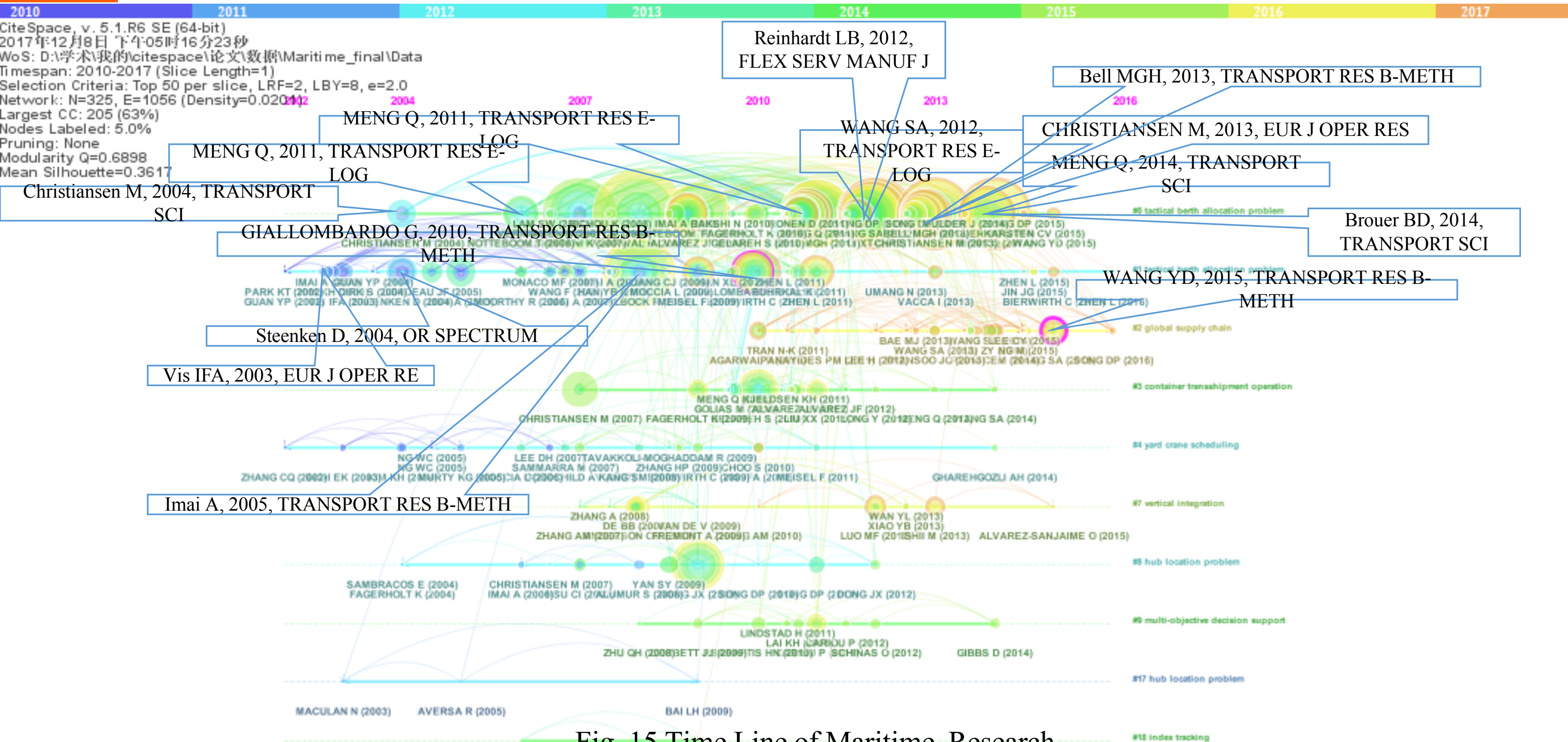


Fig. 15 Time Line of Maritime Research

3.2.2 Maritime Knowledge Basis

Key References

Table 20 Detail of the Top 10 References in Maritime Research

References	Counts	Centrality	Cluster
Agarwal, R., & Ergun, Ö. (2008). Ship scheduling and network design for cargo routing in liner shipping. <i>Transportation Science</i> , 42(2), 175-196.	37	0.04	0
Shintani, K., Imai, A., Nishimura, E., & Papadimitriou, S. (2007). The container shipping network design problem with empty container repositioning. <i>Transportation Research Part E: Logistics and Transportation Review</i> , 43(1), 39-59.	31	0.01	0
Qi, X., & Song, D. P. (2012). Minimizing fuel emissions by optimizing vessel schedules in liner shipping with uncertain port times. <i>Transportation Research Part E: Logistics and Transportation Review</i> , 48(4), 863-880.	31	0.02	0
Meng, Q., Wang, S., Andersson, H., & Thun, K. (2014). Containership routing and scheduling in liner shipping: Overview and future research directions. <i>Transportation Science</i> , 48(2), 265-280	30	0.01	0
Christiansen, M., Fagerholt, K., & Ronen, D. (2004). Ship routing and scheduling: Status and perspectives. <i>Transportation science</i> , 38(1), 1-18.	30	0.00	0
Stahlbock, R., & Voß, S. (2008). Operations research at container terminals: a literature update. <i>OR spectrum</i> , 30(1), 1-52.	30	0.04	1
Meng, Q., & Wang, S. (2011). Liner shipping service network design with empty container repositioning. <i>Transportation Research Part E: Logistics and Transportation Review</i> , 47(5), 695-708.	29	0.01	0
Wang, S., & Meng, Q. (2012). Liner ship route schedule design with sea contingency time and port time uncertainty. <i>Transportation Research Part B: Methodological</i> , 46(5), 615-633.	29	0.01	0
Bell, M. G., Liu, X., Angeloudis, P., Fonzone, A., & Hosseinloo, S. H. (2011). A frequency-based maritime container assignment model. <i>Transportation Research Part B: Methodological</i> , 45(8), 1152-1161.	29	0.08	0
Álvarez, J. F. (2009). Joint routing and deployment of a fleet of container vessels. <i>Maritime Economics & Logistics</i> , 11(2), 186-208.	27	0.04	0

NOTE : Count from all clusters

3.2.2 Maritime Knowledge Basis

Cited Journals of Key References

Table 21 Top 6 Most Cited Journals in Cited References of Maritime Research

Citation Counts	Journals
61	TRANSPORT RES E-LOG
40	EUR J OPER RES
31	TRANSPORT RES B-METH
15	TRANSPORT SCI
15	MARIT POLICY MANAG
13	OR SPECTRUM

Note : Count from all clusters

3.2.3 Aviation & Maritime Knowledge Basis

Knowledge Clusters

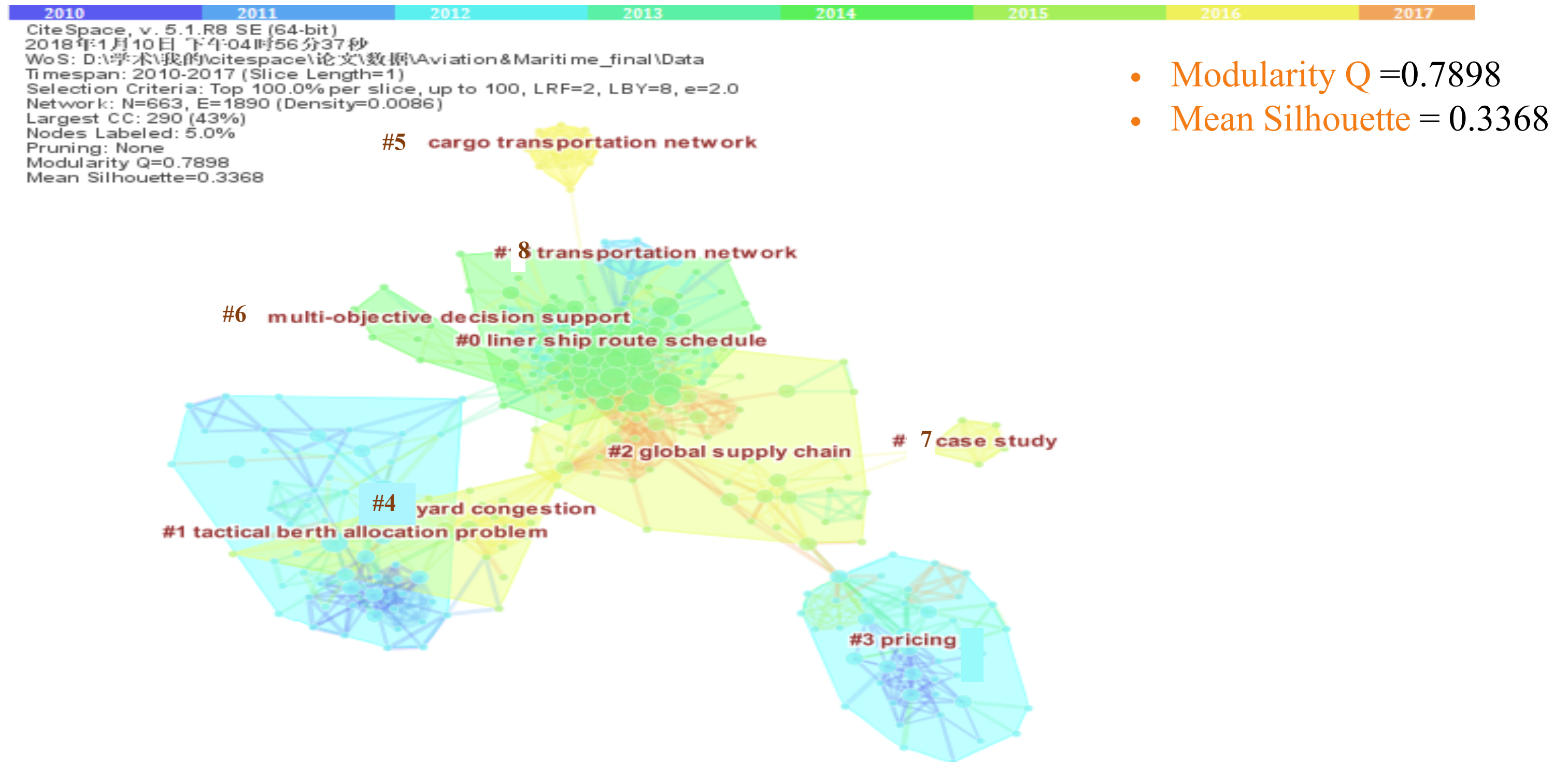


Fig.16 Co-citation Clusters of Cited References in Aviation & Maritime Research

3.2.3 Aviation & Maritime Knowledge Basis_

Knowledge Clusters

Table 22 Top terms in Each Knowledge Cluster in Aviation & Maritime Research

Cluster	Size	Sihuoette	Mean Year	LLR
0	90	0.826	2010	liner ship route schedule (2162.29, 1.0E-4); robust schedule design (1141.8, 1.0E-4); liner shipping service (1141.8, 1.0E-4); liner shipping network (533.32, 1.0E-4); container ship (533.32, 1.0E-4)
1	59	0.907	2007	tactical berth allocation problem (1772.99, 1.0E-4); yard crane scheduling (714.58, 1.0E-4); hybrid parallel genetic algorithm (714.58, 1.0E-4); greedy randomized adaptive search solution (459.82, 1.0E-4); continuous berth allocation problem (459.82, 1.0E-4)
2	47	0.765	2012	global supply chain (1112.22, 1.0E-4); ocean container transport (1112.22, 1.0E-4); research opportunities (1112.22, 1.0E-4); liner ship route schedule (485.5, 1.0E-4); robust schedule design (261.84, 1.0E-4)
3	42	0.991	2008	pricing (1122.83, 1.0E-4); airport profits matter (1122.83, 1.0E-4); slot policies (1122.83, 1.0E-4); liner ship route schedule (577.21, 1.0E-4); congestion pricing (501.43, 1.0E-4)
4	20	0.919	2011	yard congestion (381.29, 1.0E-4); yard template (381.29, 1.0E-4); container port (381.29, 1.0E-4); liner ship route schedule (71.25, 1.0E-4); robust schedule design (38.57, 1.0E-4)
5	11	0.999	2012	cargo transportation network (122.4, 1.0E-4); stochastic resource allocation (122.4, 1.0E-4); discrete stochastic demand (91.19, 1.0E-4); random network resource (91.19, 1.0E-4); standardized cargo transportation network (91.19, 1.0E-4)
6	9	0.969	2010	multi-objective decision support (252.85, 1.0E-4); future direction (252.85, 1.0E-4); environmental sustainability (252.85, 1.0E-4); liner ship route schedule (29.25, 1.0E-4); robust schedule design (15.84, 1.0E-4)
7	6	0.991	2012	case study (27.34, 1.0E-4); spatial-temporal evolution (27.34, 1.0E-4); gateways hinterland (27.34, 1.0E-4); south korea (23.84, 1.0E-4); regional economies (23.84, 1.0E-4)
8	6	0.977	2006	transportation network (68.25, 1.0E-4); hub location problem (68.25, 1.0E-4); network design (45.17, 1.0E-4); hub location (45.17, 1.0E-4); fleet deployment (45.17, 1.0E-4)

3.2.3 Aviation & Maritime Knowledge Basis

Knowledge Clusters

Table 23 The Most Active Citer in Each Cluster of Aviation & Maritime Research

Cluster	Coverage	Author and Year	Articles
0	23%	Wang, SA (2012)	Liner ship route schedule design with sea contingency time and port time uncertainty
	20%	Wang, SA (2012)	Robust schedule design for liner shipping services
	18%	Wang, SA (2012)	Sailing speed optimization for container ships in a liner shipping network
	13%	Wang, SA (2012)	Liner ship fleet deployment with container transshipment operations
1	22%	Giallombardo, G (2010)	Modeling and solving the tactical berth allocation problem
	20%	Lee, DH (2010)	The continuous berth allocation problem: a greedy randomized adaptive search solution
	19%	He, JL (2010)	A hybrid parallel genetic algorithm for yard crane scheduling
2	26%	Lee, CY (2017)	Ocean container transport in global supply chains: overview and research opportunities
3	31%	Basso, LJ (2010)	Pricing vs. Slot policies when airport profits matter
	29%	Zhang, AM (2010)	Airport capacity and congestion pricing with both aeronautical and commercial operations
	14%	Czerny, AI (2010)	Airport congestion management under uncertainty
4	55%	Zhen, L (2016)	Modeling of yard congestion and optimization of yard template in container ports
5	27%	Wang, XC (2016)	Optimal allocation of limited and random network resources to discrete stochastic demands for standardized cargo transportation networks
	27%	Wang, XC (2016)	Stochastic resource allocation for containerized cargo transportation networks when capacities are uncertain
6	67%	Mansouri, SA (2015)	Multi-objective decision support to enhance environmental sustainability in maritime shipping: a review and future directions
	11%	Wong, EYC (2015)	An utility-based decision support sustainability model in slow steaming maritime operations
7	33%	Yang, JL (2016)	Analyzing the spatial-temporal evolution of a gateway's hinterland: a case study of shanghai, china
	17%	Park, JS (2016)	The impact of seaports on the regional economies in south korea: panel evidence from the augmented solow model
8	33%	Gelareh, S (2011)	Fleet deployment, network design and hub location of liner shipping companies
	33%	Gelareh, S (2011)	Hub location problems in transportation networks

3.2.3 Aviation & Maritime Knowledge Basis

Cited Journals of Key References

Table 24 Top 7 Most Cited Journals in Cited References of Aviation & Maritime Research

Citation Counts	Journals
114	TRANSPORT RES E-LOG
65	TRANSPORT RES B-METH
51	EUR J OPER RES
35	TRANSPORT SCI
21	MARIT POLICY MANAG
19	TRANSPORT RES A-POL
15	OR SPECTRUM

Note : Count from all clusters



Outline

- 01 Knowledge Mapping Using CiteSpace
- 02 Research Method
- 03 Knowledge Mapping results
- 04 **Summary**

4. Summary

Aviation Research

- In the aviation research field, “**competition**” is a hot research topic but we seldom observe “cooperation” (Fig.6 and Table 6).
- **USA, China and Canada** have lots of research in aviation field (Fig.8 and Table 8). **University of British Columbia, University of California Berkeley and MIT** are the top 3 institutions in aviation research.(Fig.9 and Table 9).
- Some of the aviation research are coupled with **tourism** and **economic** topics (Table 16).

4. Summary

Maritime Research

- In the maritime research field, researchers mainly focus on optimization and model (Fig 10 and table 10), so most of the publications are closely related to the **OR** journals (Table 21). Meanwhile, there are also some research focusing on **finance, economic-related stream**.
- In the country cooperation network, **Singapore** has a large amount of publications, but the centrality is low. This indicates Singapore has strong research groups, but the cooperation with other countries is not frequent. (Fig.8 and Table 8).

4. Summary

Aviation & Maritime Research

- For both research areas, **no connection** between scholars from each field is observed in Fig 11 .
- The **main research topics** in aviation and maritime are **different**, and it seems that the publications in maritime dominate those in aviation.



Thank you.
Questions and Comments are welcome!